

MODEL
TED (15) 6073
REVISION –2015

Reg.No.
Signature.....

SIXTH SEMESTER DIPLOMA EXAMINATION IN CHEMICAL ENGINEERING
PARTICLE TECHNOLOGY

(Maximum. Marks::100)

Time: 3 Hrs

PART-A

Marks.

I Answer the following questions in one or two sentences. Each question carries 2 marks.

1. Define specific cake resistance.
2. Define particle size.
3. Explain sedimentation.
4. Define power number.
5. Define angle of repose.

(5x2)

PART-B

II Answer any five of the following questions. Each question carries 6 marks.

1. Draw the sketch and explain the working of top driven centrifuge.
2. Differentiate constant pressure filtration and constant rate filtration.
3. Draw the sketch of dodge jaw crusher.
4. Differentiate Rittingers law and Kicks law.
5. Explain the determination of thickner area.
6. State the difference between agitation and mixing.
7. Describe the working of a chain conveyor.

(6x5)

PART-C

(Answer one full question from each unit. Each full question carries 15 marks.)

III

MODULE-I

- (1) Draw the sketch and explain the working of a leaf filter. (8)
- (2) Explain about moore filter. (7)

OR

IV

- (1) Draw the sketch and explain the working of disc centrifuge. (8)
 - (2) Draw the sketch of super centrifuge. (7)
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MODULE-II

- V
- (1) Draw the sketch and explain the working of smooth roll crusher. (8)
 - (2) Illustrate working of trammel. (7)

OR

- VI
- (1) Compare taylor standards and U.S standards. (8)
 - (2) Draw the sketch of a Vibrating screen and explain the working. (7)

MODULE-III

- VII
- (1) Explain various stages in batch sedimentation. (8)
 - (2) Draw the sketch and explain the working of an agitation equipment.(7)

OR

- VIII
- (1) Draw the sketch of V type mixer and explain the working. (8)
 - (2) Explain the functions of draft tube and baffle in agitators. (7)

MODULE-IV

- IX
- (1) Draw the sketch and explain the working of fluidized bed catalytic cracking. (8)
 - (2) Explain working of apron conveyor. (7)

OR

- X
- (1) Explain about classification of conveyors. (8)
 - (2) Define a gas holder with a sketch. (7)
